WARNING

Detailed information on standard workshop and safety procedures, and general servicing operations is not included in this manual, which has been prepared to assist qualified service personnel. ODG assumes no responsibility or liability for PERSONAL INJURY or VEHICLE DAMAGE which results from any servicing procedure performed, including those instructions outlined in this manual. Before performing a servicing operation, an individual must have determined to his/her satisfaction that a personal injury or vehicle damage will not result from the servicing procedure or tools selected.

CAUTION

All transmission work should be performed by a qualified mechanic. Severe damage and/or reduced performance can result from an improperly serviced transmission.

CAUTION

Do not attempt to move the gearshift from the neutral starting position until the engine idles down completely. The Argo is equipped with an automatic clutch that is activated by engine speed. If the engine idle speed is too high, the transmission will grind during gear engagement.

Transmission Oil

Maintenance Schedule

The transmission oil should be changed after the initial 25 hours of vehicle operation. After this, it should be changed every 150 operating hours.

CAUTION

Proper transmission lubrication will reduce wear, and extend gearbox life. Please use the type, and quantity of transmission oil specified below.

CAUTION

Inspecting the condition and level of oil in the transmission must be done on a regular basis.

Do not operate the ARGO if the transmission oil level on the dipstick is above or below the grooved indicator. Damage to the transmission or a reduction in vehicle performance could occur.

Recommended Oil Type

Oil Type: 80W90 Gear Oil HYPOY-C or an EP Rated 80W90 Gear Oil for Extreme Pressure Gear applications.

Using any other than the recommended oil could cause serious damage to the transmission components. The approximate refill capacity of the ARGO transmission is 0.85 Liters (1.5 imperial pints, 1.8 U.S pints). However, be sure that it is filled to the full mark on the dipstick. DO NOT OVERFILL.

Do not operate the ARGO if the transmission oil level on the dipstick is above or below the grooved indicator. Damage to the transmission or a reduction in vehicle performance could occur.

Recommended Oil Type

Oil Type: 80W90 Gear Oil HYPOY-C or an EP Rated 80W90 Gear Oil for Extreme Pressure Gear applications or ARGO brand 75W90 Synthetic lube (Part No.130-104).

Using any other than the recommended oil could cause serious damage to the transmission components. The approximate refill capacity of the ARGO transmission is 0.85 Liters (1.5 imperial pints, 1.8 U.S pints). However, be sure that it is filled to the full mark on the dipstick. DO NOT OVERFILL.

Oil Level Inspection

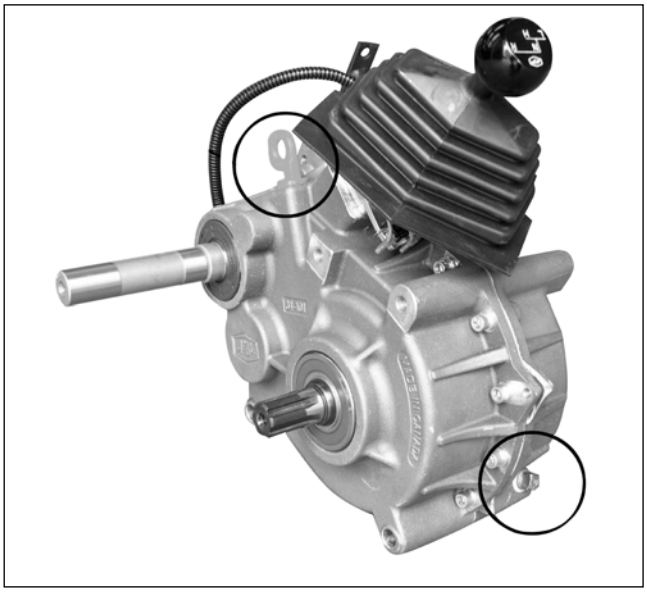
Inspect the transmission oil on a daily basis.

CAUTION

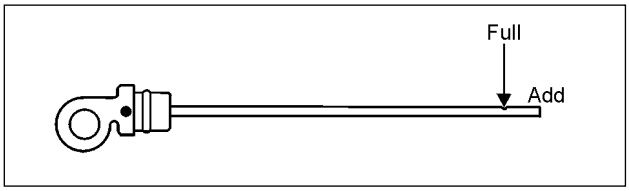
Do not run the ARGO if the oil level in the transmission shows above or below the mark on the dipstick. Damage to transmission components could occur.

Checking the Transmission Oil Level (34-100)

Check the transmission oil level every 50 operating hours. Some models of the ARGO are equipped with a transmission oil dipstick Clean the area around the dip-stick before removing. Remove the dipstick by pulling up.

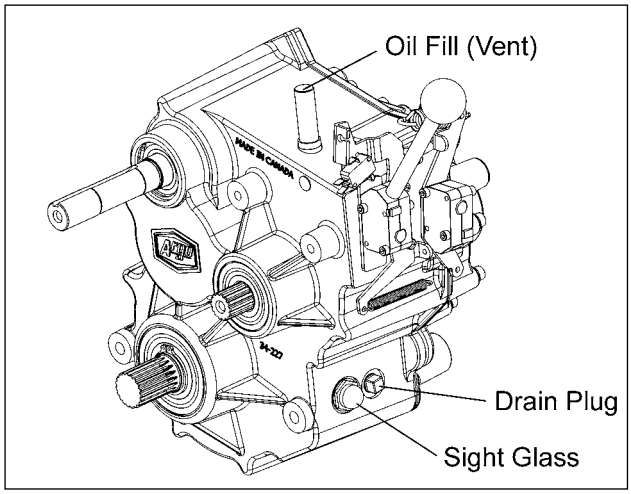


The transmission oil level should be even with the mark on the dipstick. Add oil through the transmission oil fill/dipstick hole until the transmission is filled to the correct level. DO NOT OVERFILL. Replace the dipstick securely.



Checking the Transmission Oil Level (34-200)

HDi (Admiral) transmission models do not have an oil dipstick. Check for correct oil level by viewing the site glass installed to the lower portion of the transmission housing. To view this site glass, remove the quick release firewall. Oil filling half the site glass indicates correct oil level.



Changing the Transmission Oil - 34-100

Remove firewall to access the drain plug located at the bottom of the transmission. Drain the transmission oil into a suitable container and dispose of the oil at a disposal site.

While draining the transmission oil, be sure to clean off any metal particles that are on the magnet of the drain plug. These fine metal particles are a result of the transmission gears meshing during the initial break-in period. Re-install the plug and tighten it securely.

Refilling the Transmission - 34-100 "Classic"

Refill the transmission through the oil fill/dipstick hole. Oil capacity for all transmissions is 38.7 oz.(1.1 L) of 80 W 90 Gear Lube HYPOY-C. As you refill the transmission, check the oil level with the dipstick. Check that the oil level is even with the mark on the dipstick, after it has been seated fully in the dipstick hole. DO NOT OVERFILL.

Overfilling may result in oil being forced out the breather hole that could contaminate the brake pads and lead to brake failure.

Changing the Transmission Oil - 34-200 "ADMIRAL"

Removing the oil from the Admiral transmission requires the use of a vacuum style pump such as the 638-02 Big Boy, Top Sider (available from Ontario Drive and Gear). Due to the design of the Admiral transmission, the majority of the oil in the case will be below the drain plug.

Remove the drain plug and drain the oil until the flow stops. Insert the vacuum tube of the Big Boy Vacu-Pump into the drain plug hole and remove the remaining oil from the transmission sump.

Refilling the Transmission - 34-200

Install the drain plug. Remove the fill/vent plug located on the top of the transmission. Fill the transmission until the sight glass is half full. For Oil capacity see Oil Capacity Chart

Transmission Oil Capacity	
Admiral (34-200)	1.2 L
Classic (34-100)	1.1 L

Removing the Transmission (34-100)

1. Disconnect all wiring to the battery before proceeding with the following component removal.
2. Perform servicing procedure, **Removing the Firewall.**
3. Perform the servicing procedure, **Remove the Front Floor Pan.**
4. Perform the servicing procedure, **Remove the Drive Belt.**
5. Perform the servicing procedure, **Remove the Driven Clutch.**
6. Perform the servicing procedure, **Remove the Hydraulic Brake Calipers - Frontier Steering Brakes.** Do NOT disconnect the brake lines.
7. Perform the servicing procedure, **Removing The Idler Chain- Frontier (Manual or Self Adjust).**
8. Loosen the fasteners that secure the brake cooling duct and slide the duct to allow removal of brake discs (Vehicles equipped with brake cooling system only).
11. Perform the servicing procedure, **Removing The Steering Brake Discs - Frontier**
12. Locate, unfasten and remove the (2) bolts, lock washers and nuts that secure the transmission to the power pack frame.
13. Tilt the transmission back towards the drivers compartment and lift it from the power pack frame.

Argo Transmission Disassembly - 34-100

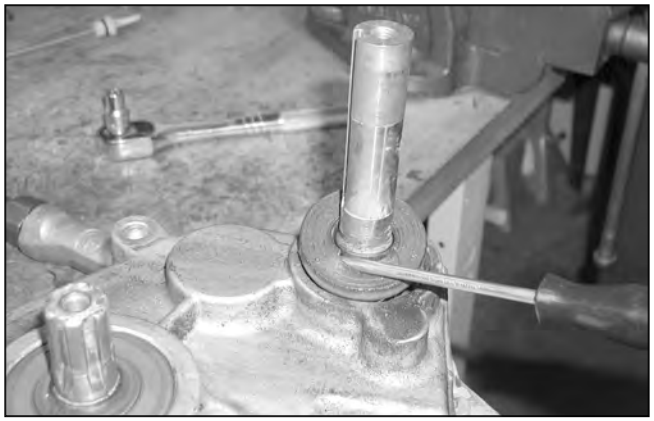
1. Drain oil and unthread the 4 fasteners securing the shifter lever assembly to top end of the transmission and remove the lever.





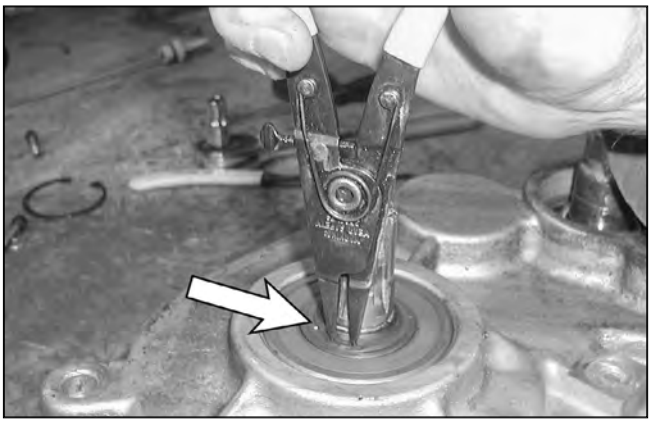
2. Remove all fasteners securing the housing halves together. There are 11 located on the input shaft side and 4 located on the opposite side. Ensure that all fasteners have been removed before proceeding to separate the transmission housings.

3. Remove the oil seal from the input shaft.
4. Remove the retaining ring from the input shaft.



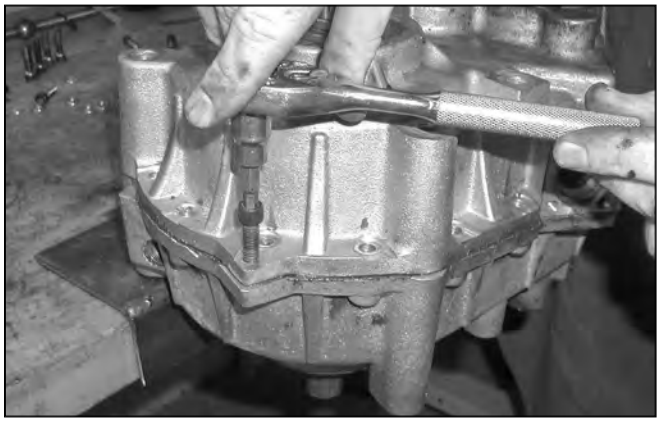


5. Remove the retaining rings from both output shafts. There may or may not be shims below the retaining rings.



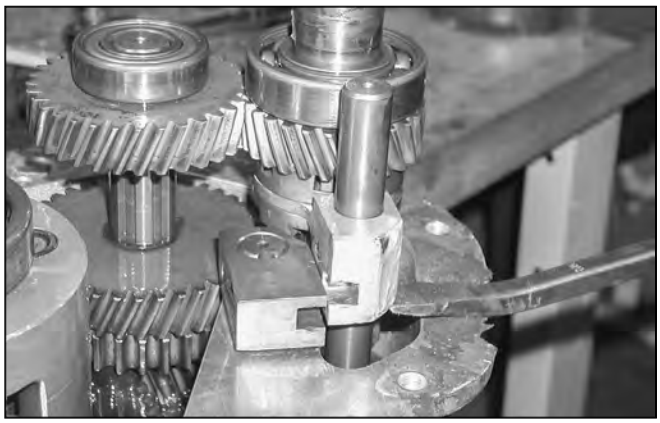
Separating the Transmission

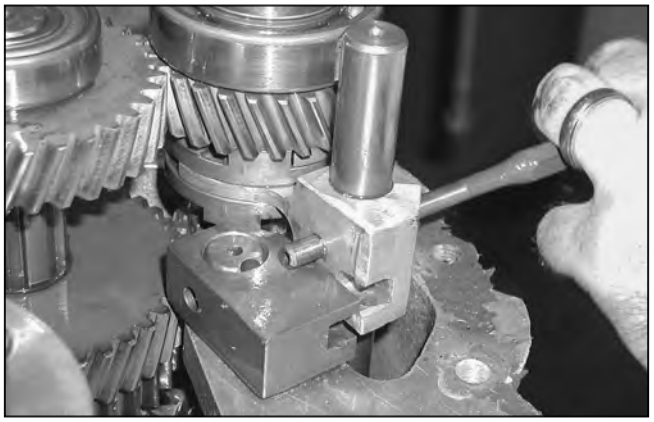
1. There are 3 threaded tab locations provided on the 34-100 transmission housing (input shaft side). These locations are used to assist in separating the two halves of the transmission during disassembly. As 3 bolts are threaded into the housing, they come into contact with the opposite housing, pushing and separating the 2 halves. Thread each of these bolts a little at a time and in an uniform alternating pattern. **DO NOT** overtighten to avoid the risk of breaking the housing tab. Stop occasionally to tap the output shaft with a soft faced hammer. Repeat this until the cover has separated.
2. Pull the housing halve, up and off from the input shaft side.



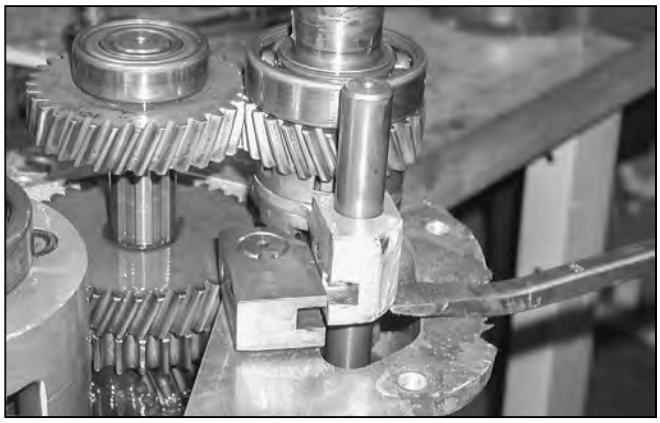


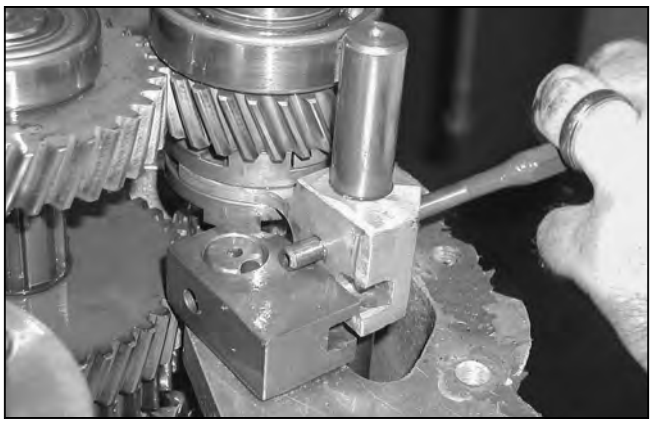
3. Shift manually to top position with a pry bar (High Gear) and remove roll pin from the shift block on the high/low shift shaft.



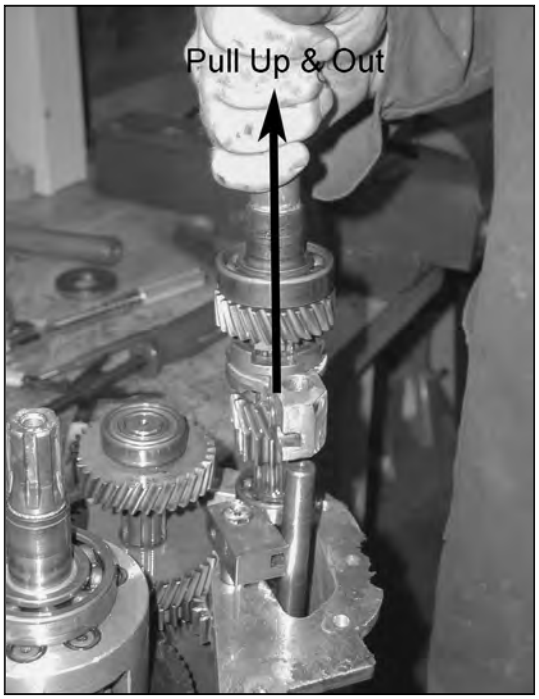


4. Place shift shaft back into neutral.
5. Place reverse shift shaft into reverse.

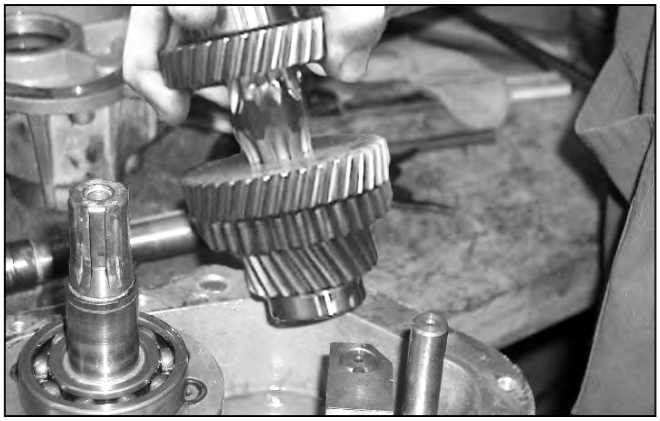


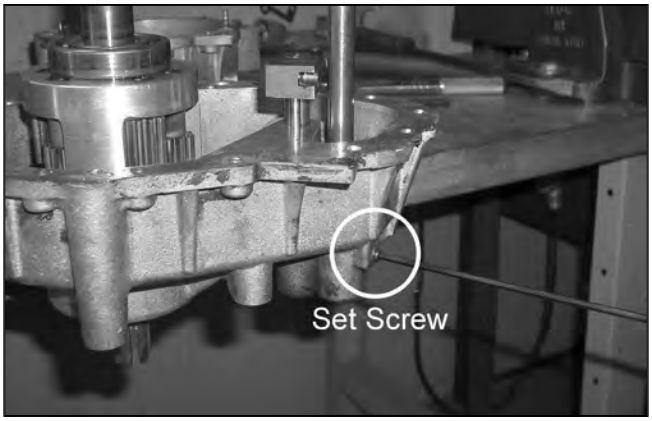


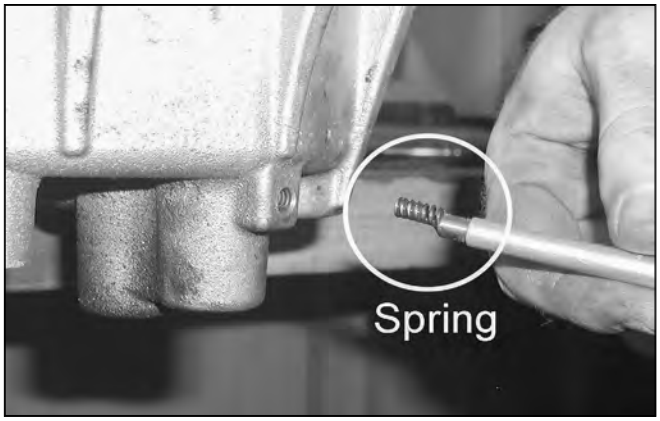
6. Pull the input shaft from the transmis sion housing.
7. Remove the spline shaft drive gear assembly. *See following page.*
8. Remove the set screw securing the ball and spring within the housing at the hi/lo shift shaft assembly. *See following page.*
9. Remove the spring and steel ball. *See following page.*

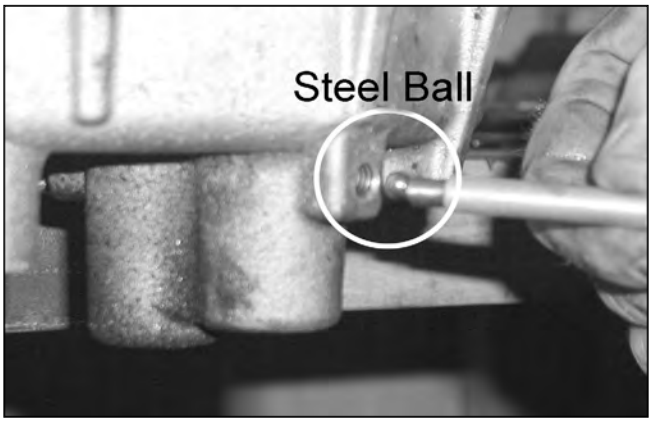


10. Place the reverse shift shaft block back down against the housing (out of reverse).
11. Remove the hi/lo shift shaft from the housing.







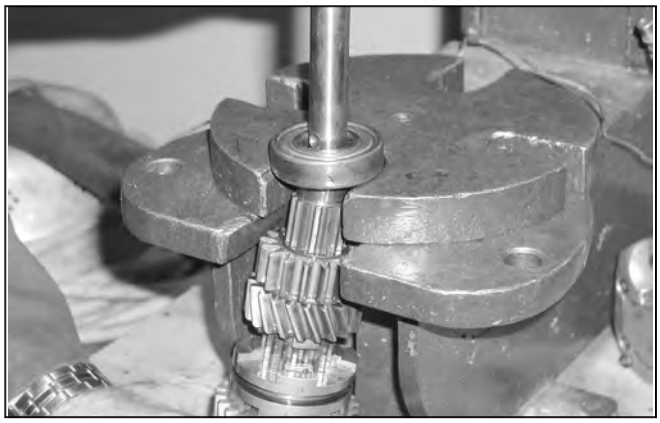


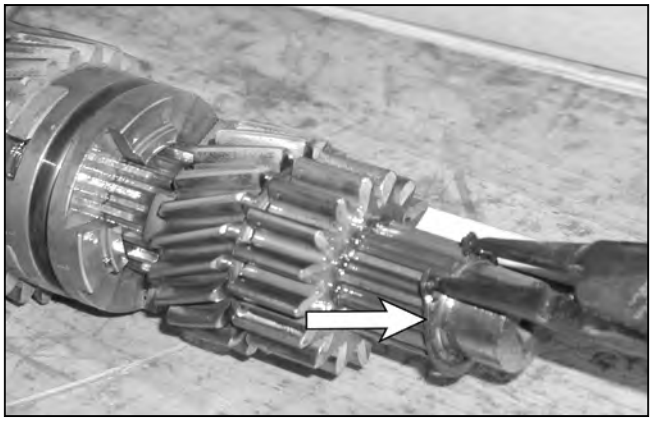




Disassembling The Input Shaft

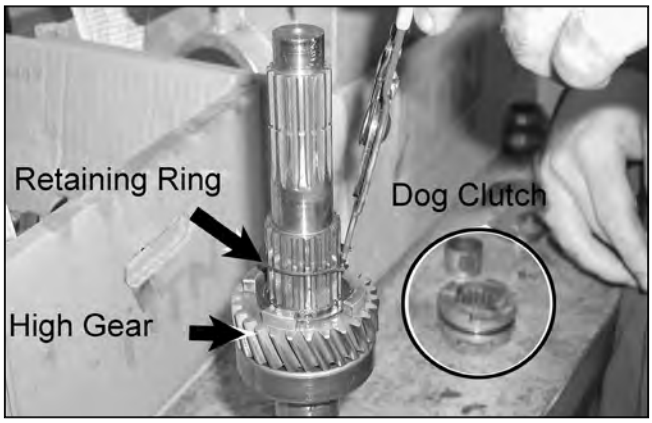
1. Using an arbor press, remove the smaller ball bearing from the end of the input shaft.
2. Remove the retaining ring at the reverse gear.



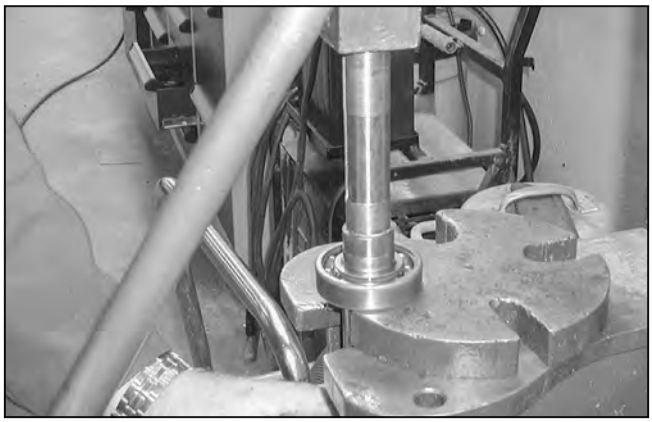


3. Remove the reverse and low gear from the input shaft at the arbor press.
4. Slip the dog clutch from the input shaft and remove the retaining ring & Thrust Washer found above the High Gear.
5. Slip the high gear from the input shaft, and remove the thrust washer and Retaining Ring found below it.
6. Remove the last ball bearing from the input shaft at the arbor press.



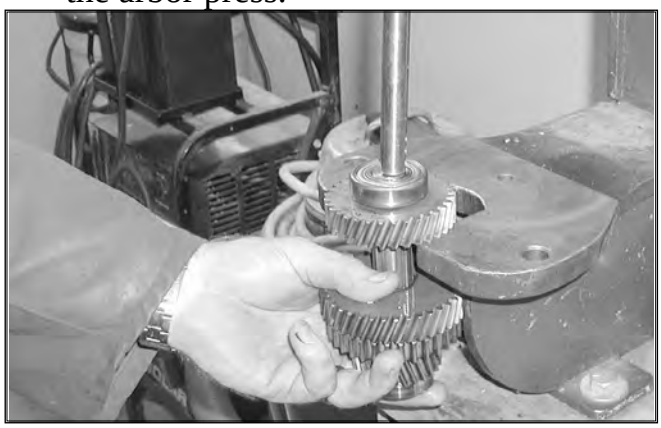
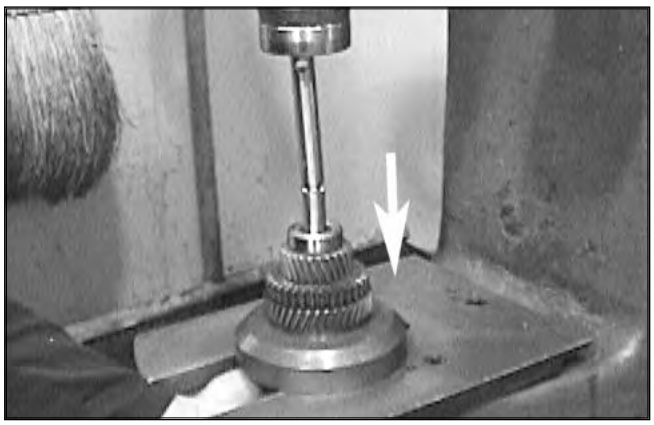






Disassemble the Spline Shaft Drive Gear Assembly

1. Remove the bearing and high gear from the spline shaft at the arbor press.
2. Remove low gear, reverse gear, drive gear and bearing from the spline shaft using the the arbor press.

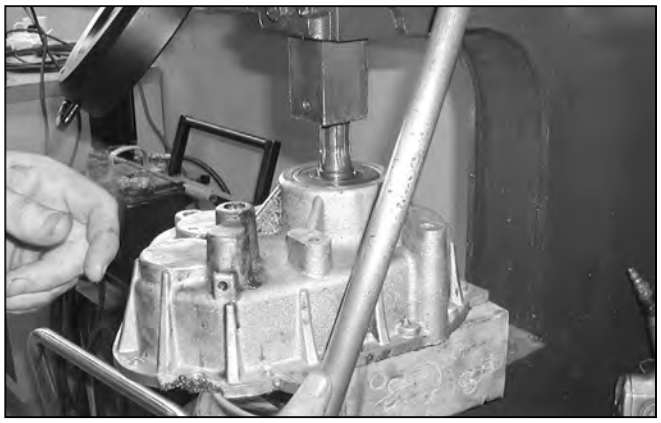
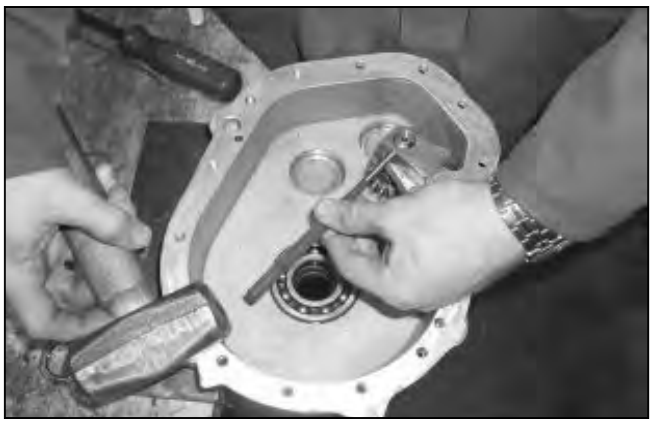
Disassembling The Housing Components

1. Press the differential from the housing at the arbor press.

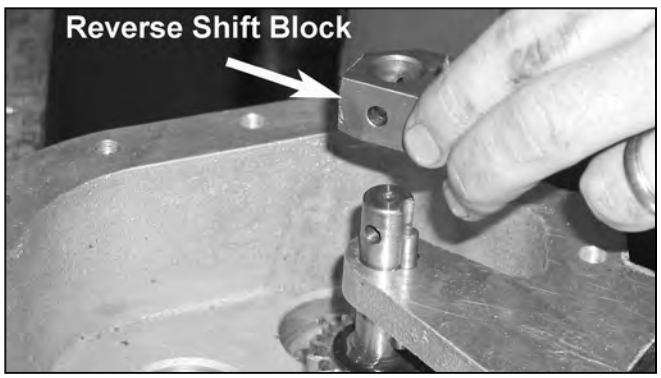
NOTE

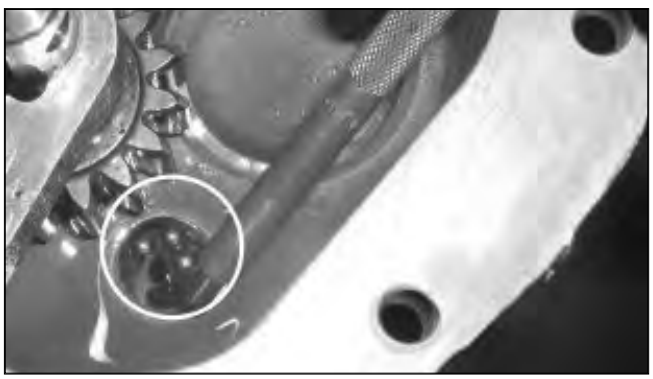
The use of wood blocks beneath the housing prevents damage to the sealing edge of the casting as well as provides height for removing the differential.

2. Disassemble reverse shift shaft block by removing the roll pin securing it to the shaft.

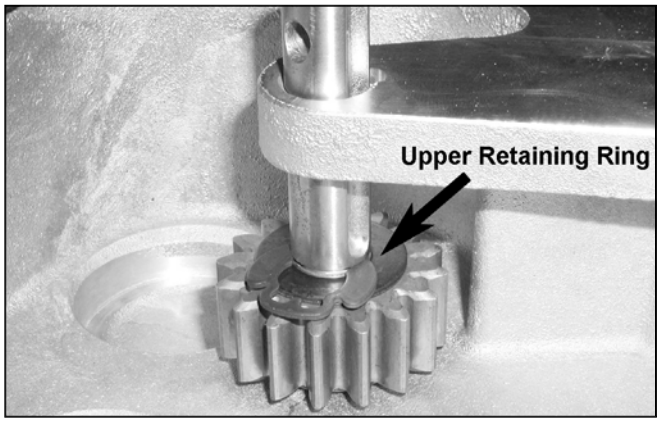



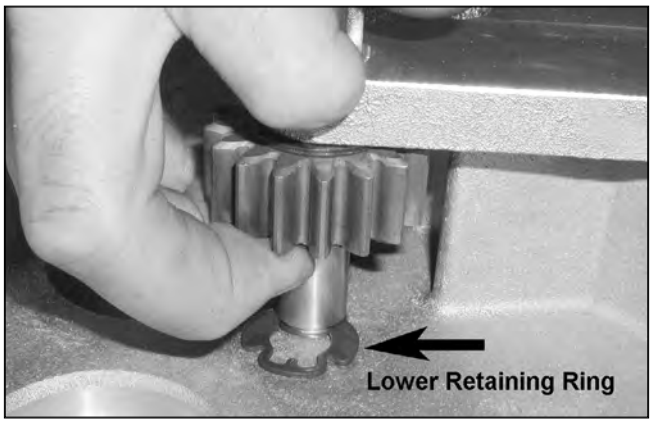
3. Remove the 3 steel balls located in the housing, between the hi/lo shift shaft and the reverse shift shaft.





4. To remove the reverse shift shaft and reverse idler gear from the housing, first remove the retaining clip located above the reverse idler.
5. Raise the reverse idler up and remove the retaining clip found below.

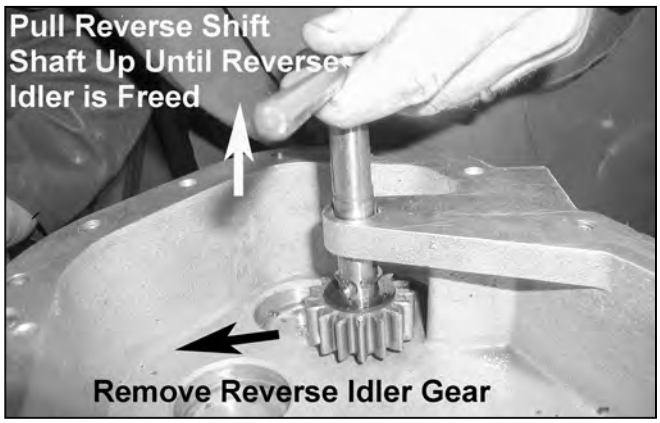


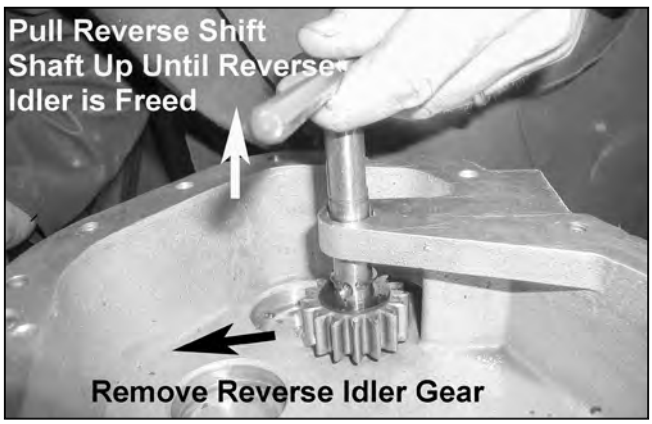


6. Pull the reverse shift shaft up and free from the housing.
7. Remove the Steel Dowel Pin from the reverse shift shaft.

IMPORTANT

There is 1 Thrust Washer to the top side of the reverse idler gear and a Retaining Thrust Washer to the underside of the reverse idler gear



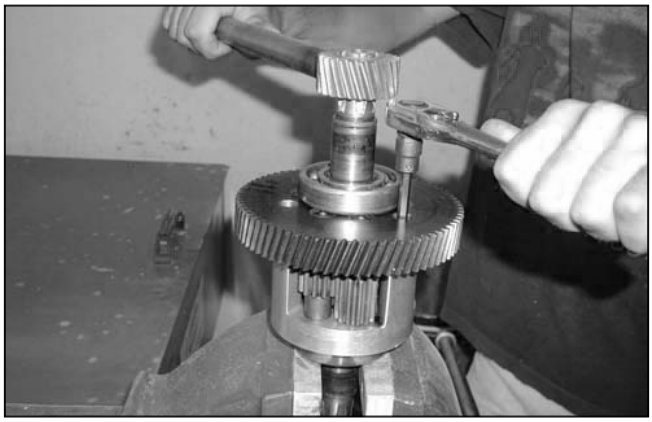


8. Remove the reverse shift shaft from the housing.
9. Remove the last ball and spring from the housing.
10. Wash and soak all components in a degreasing bath before inspection.



Disassembling The Differential

1. Secure the Differential by the Output shaft on the cage side within a vise. A vise equipped with aluminum jaws must be used to avoid any damage to the out-put shaft.
2. Hold the Output Shaft from turning while removing the 3 Fasteners that hold the Gear Plate and Aluminum Cage together.

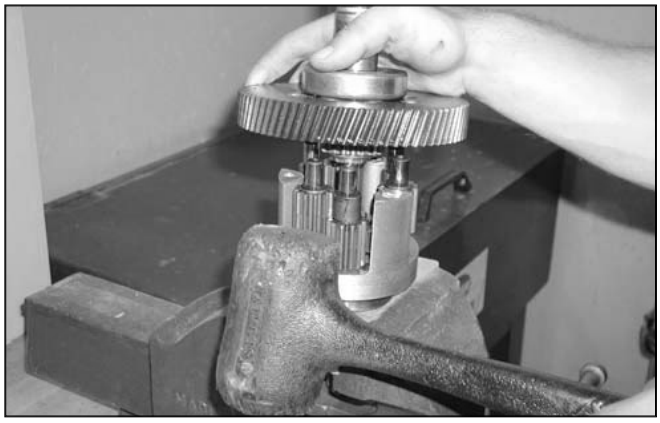
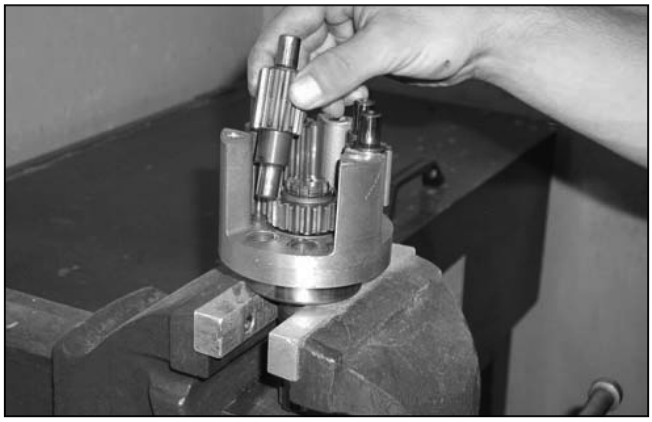



3. With the Fasteners removed, gently tap the Gear Plate from the Aluminum Cage using a soft faced mallet.

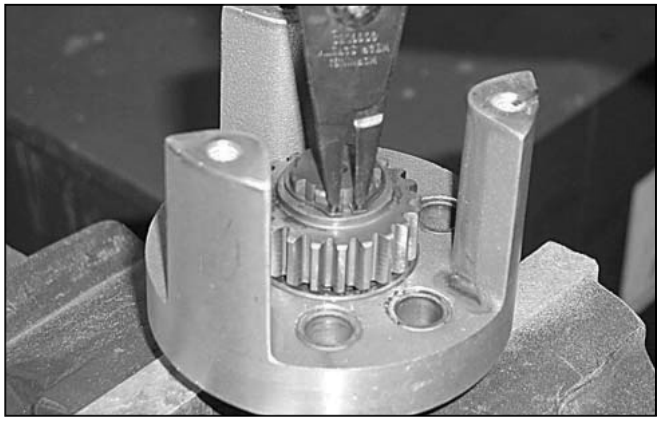
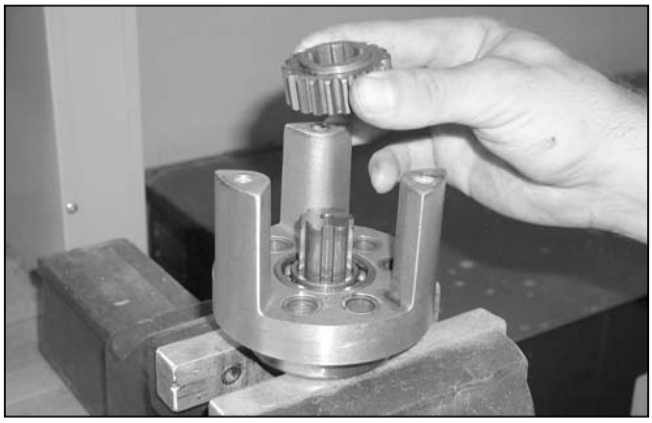
CAUTION

When lifting the Gear Plate from the Cage, be aware of any Pinions that may come with the Gear Plate. Avoid any of these falling to the floor. Any damage to these pinions can result in reduction of gear life or noise issues.

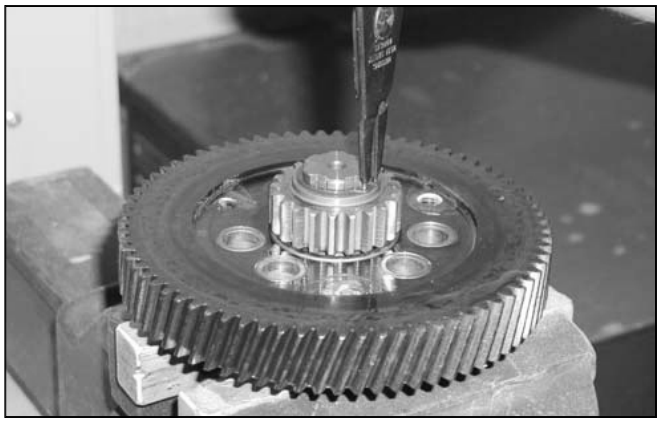
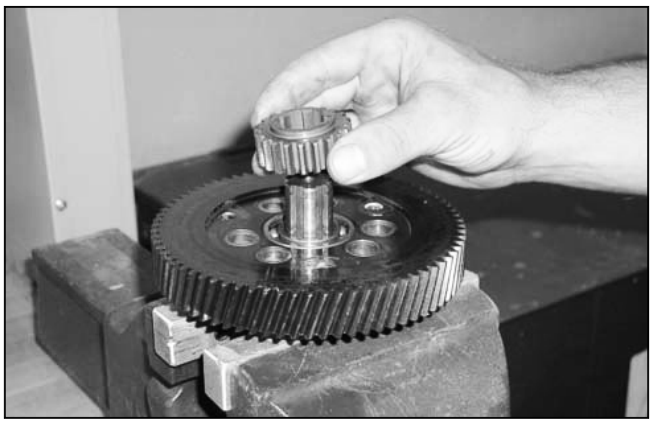
4. Remove each of the (6) Pinions.

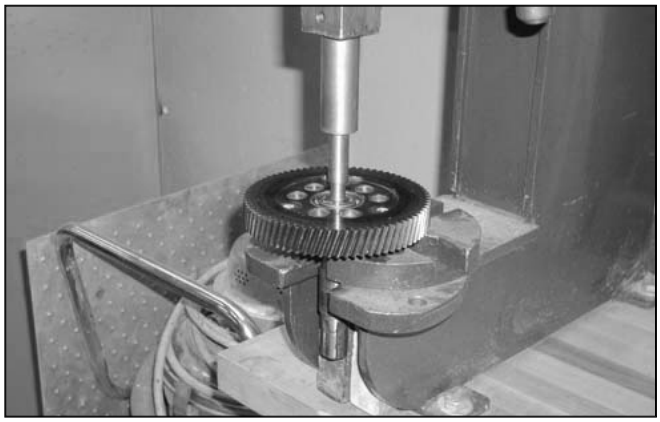
5. Remove the Retaining Ring that secures the Sun Gear to the Output Shaft on the Aluminum Cage side. *See following page*.
6. Slip the Sun Gear from the Output Shaft. *See following page.*

7. Remove the Retaining Ring that secures the second Sun Gear to the Output Shaft on the Gear Plate side. *See following page*.
8. Slip the Sun Gear from the Output Shaft Gear Plate side.

9. Press the Output Shafts from both Cage and Gear Plate.




10. Remove each of the 101-15 Bearings from both Cage and Gear Plate. Using the 608-54 Bearing Removal Tool, remove each bearing from both Gear Plate and the Aluminum Cage. Tap with a soft faced mallet. *See following page.*



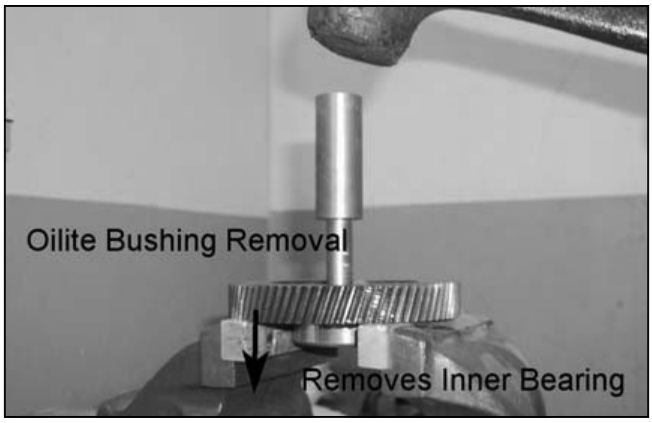


11. Remove all (12) Oilite Bushings from both Cage and Gear Plate using 658-05 Needle Bearing Remover/Installer.

NOTE

This procedure will also push out the Output Shaft Bearing that has more than likely remained assembled to both of these components.





Transmission Component Inspection

Wash all parts in a clean varsol bath and dry all parts with a lint free rag. Lay out all parts for inspection.

Mechanical transmissions do not need a great deal of maintenance beyond periodic and regular oil changes. However, once a problem has been detected and the transmission has been disassembled, it is important to examine all transmission components for wear, cracks or breakage. Repairing or replacing all faulty components will prevent further premature breakdowns and the need to disassemble the transmission again in the near future.

Inspect all transmission components and look for: Broken or badly worn teeth

- worn out or seized up bearings
- badly worn spline shafts
- cracks in the transmission housing or cover
- cracks in aluminum differential cage

IMPORTANT

Lack of lubrication, wrong type of lubrication or transmission overheating may cause excessive wear or damage to transmission gears. When inspecting gears, follow these guidelines.

Normal Wear

As gears operate, the meshing operation and oil wash will polish the gear teeth, removing any sharp edges, imperfections or shards. The polished surface should cover the entire length and depth of the teeth. Well lubricated gears will show this condition after many hours of operation.

The initial 25 hrs of operation on a new transmission will indicate this process when the oil is changed for the first time. The oil will appear very silvery and fine metal filings will be noticeable on the drain plug's magnet.
Abrasive Wear and Scratching

Metal particles which flake off gears, shafts and castings are carried in the lubricant and can cause uneven and premature wear to gear teeth. Thorough cleaning of all components at disassembly and fresh transmission oil can normally prevent further wear of this kind.

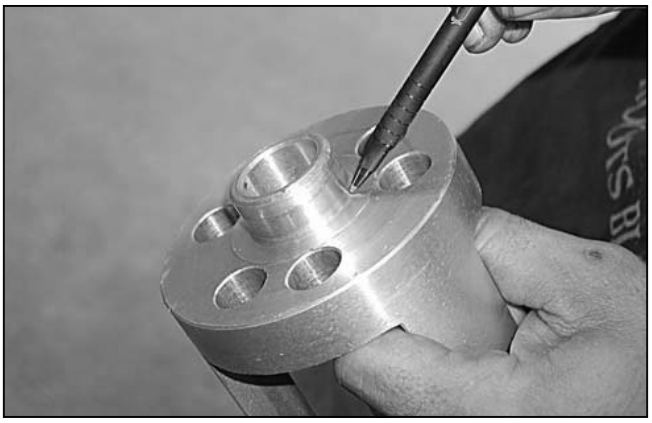
Breakage

Any gear or spline shaft with cracked, broken or missing teeth should be replaced immediately. The installation of a badly damaged gear or shaft will lead to premature wear of other components and other potential breakdowns.

Replace all badly worn, broken or missing components immediately with genuine ARGO replacement parts.

If there is any doubt as to whether the transmission component needs replacing, generally the best thing to do is to replace it while the job is at hand.

Aluminum Cage Inspection



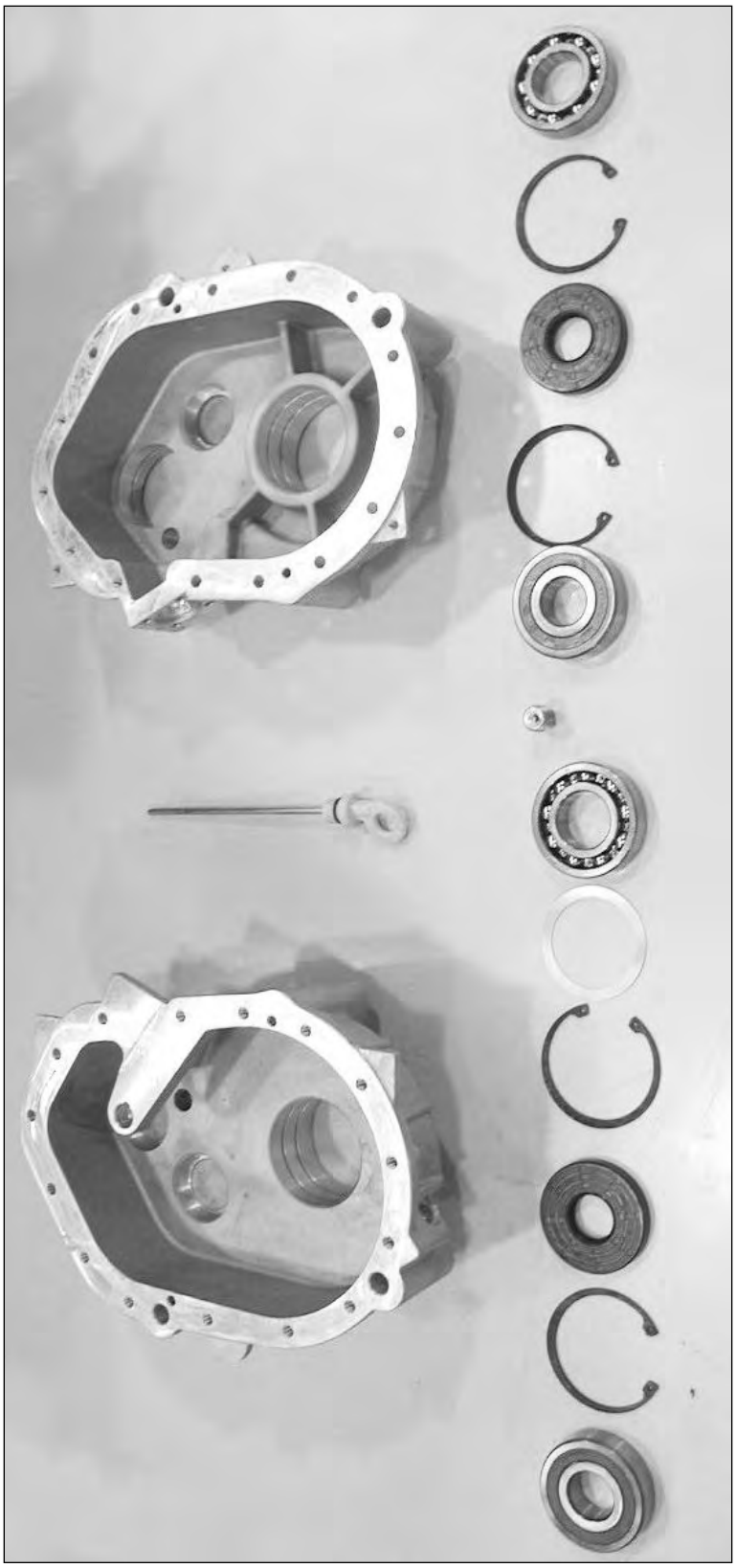
Inspect the aluminum cage for any hairline cracks, especially at the shoulder where the inner bearing is seated.

Bearings

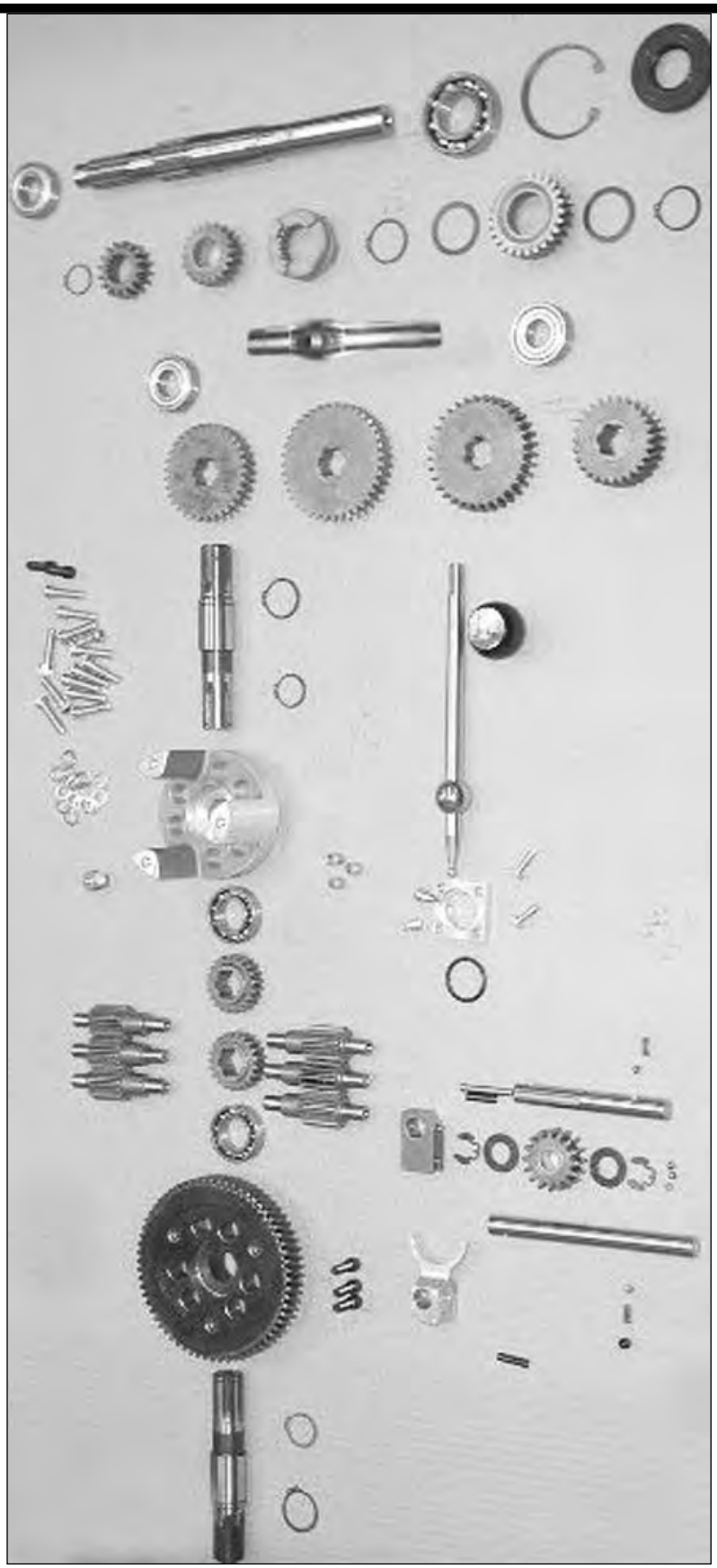
Unless the ARGO technician has determined to his/her own satisfaction that the bearings are suitable to reuse, it is recommended that all bearings be replaced when rebuilding a transmission.

Reassembling the Transmission

1. Lay out all components of the 34-100 "Classic" Transmission on a clean work surface.



34-100 Transmission "Classic"



34-100 Transmission "Classic"

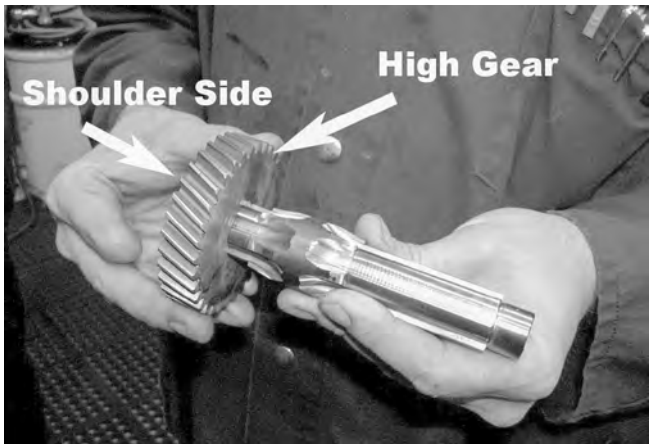
Assembling the Idler Gear Bore Assembly

1. Install the 34-109 High Gear to the short side of the 34-112 Splined Shaft.

IMPORTANT

Install the High Gear with the machined shoulder up.

2. Press the 101-01 Bearing to the High Gear end of the Splined Shaft with the shielded side facing out.



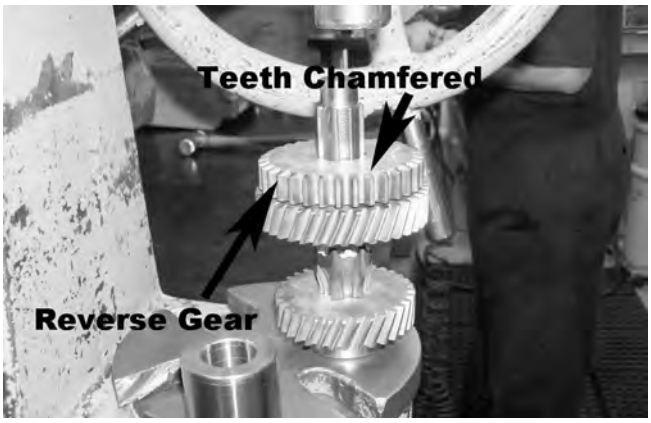


3. Install the 34-110 Low Gear to the opposite side of the Splined Shaft (long side).

IMPORTANT

Install the Low Gear with the machined shoulder up.





IMPORTANT

The 34-111 Reverse gear must be assembled to the Splined Shaft with the chamfered teeth side facing up.